|INCH-POUND|
MIL-D-83531/1A
23 Nov 1992
SUPERSEDING
MIL-D-83531/1
31 March 1987

MILITARY SPECIFICATION SHEET

DELAY LINES, 10-TAP, DUAL-IN-LINE, 14-PIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and standards (DODISS) specified in the solicitation: MIL-D-83531.

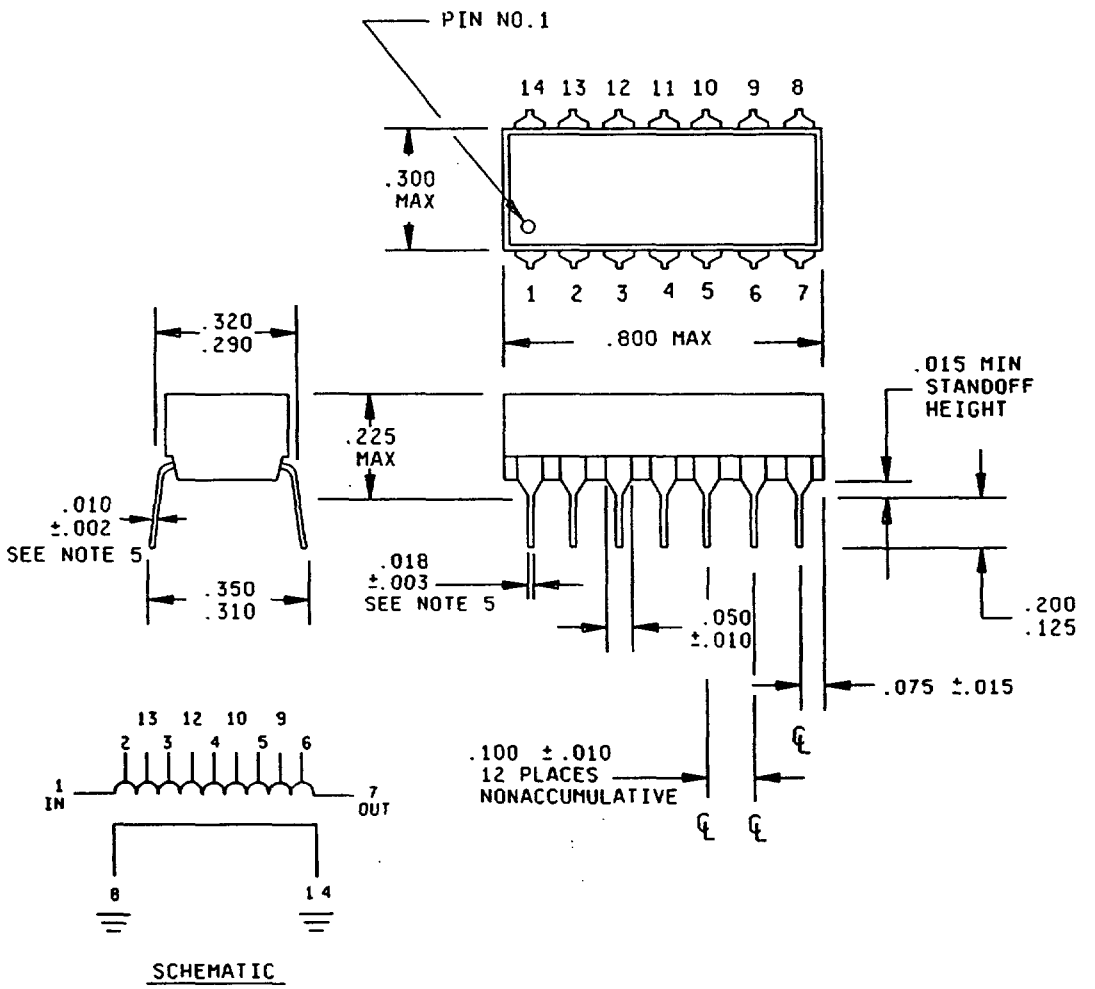


FIGURE 1. Case and mounting dimensions.

(A) denotes changes

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Inches	mm
.002	0.05
.003	0.08
.010	0.25
.015	0.38
.018	0.46
.050	1.27
.075	1.91
.125	3.18
.200	5.08
.225	5.72
.290	7.37
.300	7.62
.310	7.87
.320	8.13
.350	8.89
.800	20.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Location and shape of standoffs are optional. Standoff height shall be as indicated.
4. Pin 11 is not connected.
5. Terminal dimensions are prior to hot solder dip.
6. Schematic is for general information only.
(A) 7. Leads shall be free of case meniscus and other foreign material and shall be solderable for a minimum of .010 above the seating plane of the delay line.

FIGURE 1. Case and mounting dimensions - Continued.

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REQUIREMENTS:

Operating temperature: -55°C to +125°C.

Weight: 2.5 grams maximum.

Case material: Metal or plastic.

(A) Terminal material: In accordance with MIL-D-83531.

Design and construction: See figure 1.

Total delay time: See table I.

Rise time: 25 percent of total delay, maximum.

Voltage attenuation: See table I.

(A) Distortion: ±15 percent at maximum delay.

Delay time variation with temperature: 100 ppm/°C maximum.

DC resistance (RDC):

$$RDC = impedance\left(\frac{attenuation}{1 - attenuation}\right)$$

$$EXAMPLE: \quad RDC = 50\left(\frac{0.03}{(1-0.03}\right) \quad RDC = 1.5\ ohms$$

Nominal characteristic impedance: See table I.

Insulation resistance: In accordance with MIL-D-83531.

Input pulse:

Peak pulse voltage: 1 to 3 volts ±5 percent.

Pulse width: Three times nominal total delay.

Duty cycle: 25 percent.

(A) Salt spray (corrosion) (metal cases only): In accordance with MIL-D-83531.

Thermal shock: In accordance with MIL-D-83531.

Shock: In accordance with MIL-D-83531.

Tap accuracy and total accuracy:

Total delay 0 to 10: ±0.5 ns.
Total delay 11 to 25: ±1 ns or 5 percent (whichever is greater).
Total delay 26 and up: ±2 ns or 5 percent (whichever is greater).

(A) Part or Identifying Number (PIN): The PIN shall be as follows:

EXAMPLE:

M83531	/01	-001
Military designator and basic specification	Specification sheet number (note zero)	Dash number (see table I)

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(A) TABLE I. PIN's and electrical characteristics.

PIN M83531/01- Nominal characteristic impedance (±10 percent)					1/ Total delay time (ns)	Maximum attenuation by impedance (percent)				
50Ω	100Ω	200Ω	360Ω	500Ω		50Ω	100Ω	200Ω	360Ω	500Ω
001	101	201	301	401	5	6	5	5	5	7
002	102	202	302	402	10	"	"	"	"	"
003	103	203	303	403	15	"	"	"	"	"
004	104	204	304	404	20	"	"	"	"	"
005	105	205	305	405	25	"	"	"	"	"
006	106	206	306	406	30	"	"	"	"	"
007	107	207	307	407	35	"	"	"	"	"
008	108	208	308	408	40	"	"	"	"	"
009	109	209	309	409	45	"	"	"	"	"
010	110	210	310	410	50	"	"	6	"	"
011	111	211	311	411	55	"	"	"	"	"
012	112	212	312	412	60	"	"	"	"	"
013	113	213	313	413	65	"	"	"	"	"
014	114	214	314	414	70	"	"	"	"	"
015	115	215	315	415	75	8	"	"	"	"
016	116	216	316	416	80	"	"	"	"	"
017	117	217	317	417	85	"	"	"	"	"
018	118	218	318	418	90	"	"	"	"	"
019	119	219	319	419	95	"	"	"	"	"
020	120	220	320	420	100	"	8	"	"	"
021	121	221	321	421	110	10	"	"	7	8
022	122	222	322	422	120	"	"	"	"	"
023	123	223	323	423	130	"	"	"	"	"
024	124	224	324	424	140	"	"	"	"	"
025	125	225	325	425	150	"	"	8	"	"
026	126	226	326		160	15	10	"	"	"
027	127	227	327		170	"	"	"	"	"
028	128	228	328		180	"	"	"	"	"
029	129	229	329		190	"	"	"	"	"
030	130	230	330		200	"	"	"	"	"
031	131	231	331		210	"	"	"	"	"
032	132	232	332		220	"	"	"	"	"
033	133	233			230	"	"	"	"	"
034	134	234			240	"	"	"	"	"
035	135	235			250	"	"	10	"	"
036	136	236			260	"	15	"	"	"
037	137	237			270	"	"	"	"	"
038	138	238			280	"	"	"	"	"
039	139	239			290	"	"	"	"	"
040	140	240			300	"	"	"	"	"
041	141	241			310	20	"	"	10	10
042	142	242			320	"	"	"	"	"
043	143	243			330	"	"	"	"	"
044	144	244			340	"	"	"	"	"
045	145	245			350	"	20	15	"	"
046	146	246			360	"	"	"	"	"
047	147	247			370	"	"	"	"	"
048	148	248			380	"	"	"	"	"
049	149	249			390	"	"	"	"	"
050	150	250			400	"	"	"	"	"
051	151				410	"	"	"	"	"
052	152				420	"	"	"	"	"
053	153				430	"	"	"	"	"
054	154				440	"	"	"	"	"
055	155				450	22	"	"	"	"
056	156				460	"	"	"	"	"
057	157				470	"	"	"	"	"
058	158				480	"	"	"	"	"
059	159				490	"	"	"	"	"
060	160				500	"	"	"	"	"

1/ Delay between successive taps shall be 1/10 of the nominal total delay time.

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QUALITY ASSURANCE PROVISIONS:

Extent of qualification: The extent of qualification shall be as specified in MIL-D-83531, for example:

Qualification and testing of M83531/01-001 and M83531/01-425 shall be sufficient to grant qualification approval to M83531/01-001 through M83531/01-425 inclusive.

(A) Supersession data: MIL-D-83531/1 supersedes MIL-D-23859/23 when qualified sources are available. MIL-D-23859/23 was inactivated for new design 12 October 1990.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5999-0301)

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